

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279021

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Huang; Yu-Chia et al.

ELECTRONIC DEVICE

Abstract

The disclosure provides an electronic device including a display panel, a data signal driving circuit, and a scan signal driving circuit. The display panel has a display area and a non-display area adjacent to the display area. The display panel includes a pixel disposed in the display area; a data connection line disposed in the non-display area; and a scan connection line disposed in the non-display area. The data connection line is electrically connected between the pixel and the data signal driving circuit. The scan connection line is electrically connected between the pixel and the scan signal driving circuit. The data connection line and the scan connection line in the non-display area are at least partially overlapped.

Inventors: Huang; Yu-Chia (Miaoli County, TW), Wu; Yuan-Lin (Miaoli County, TW), Tsai; Tsung-Han (Miaoli County, TW), Lee; Kuan-Feng (Miaoli County, TW)

Applicant: Innolux Corporation (Miaoli County, TW)

Family ID: 80446654

Assignee: Innolux Corporation (Miaoli County, TW)

Appl. No.: 19/208605

Filed: May 15, 2025

Foreign Application Priority Data

CN

202110161626.0

Feb. 05, 2021

Related U.S. Application Data

parent US continuation 18596676 20240306 parent-grant-document US 12333980 child US 19208605

parent US continuation 18164557 20230203 parent-grant-document US 11961444 child US 18596676

Publication Classification

Int. Cl.: G09G3/20 (20060101)

U.S. Cl.:

CPC G09G3/20 (20130101); G09G2300/0404 (20130101); G09G2310/0275 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 18/596,676, filed on Mar. 6, 2024. The prior application Ser. No. 18/596,676 is a continuation application of and claims the priority benefit of a prior application Ser. No. 18/164,557, filed on Feb. 3, 2023. The prior application Ser. No. 18/164,557 is a continuation application of and claims the priority benefit of a prior U.S. application Ser. No. 17/571,531, filed on Jan. 10, 2022, which claims the priority benefit of China application serial no. 202110161626.0, filed on Feb. 5, 2021. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a transparent display device.

Description of Related Art

[0003] With the development of related display technologies, display devices have been applied to many products. In order to match the functions and characteristics of a product with the product itself, transparent display devices are required in many designs. Therefore, research and development of transparent display devices have gradually attracted attention.

SUMMARY

[0004] The disclosure provides a transparent display device.

[0005] According to embodiments of the disclosure, an electronic device including a display panel, a data signal driving circuit, and a scan signal driving circuit. The display panel has a display area and a non-display area adjacent to the display area. The display panel includes a pixel disposed in the display area; a data connection line disposed in the non-display area; and a scan connection line disposed in the non-display area. The data connection line is electrically connected between the pixel and the data signal driving circuit. The scan connection line is electrically connected between the pixel and the scan signal driving circuit. The data connection line and the scan connection line in the non-display area are at least partially overlapped.

[0006] Based on the above, in the embodiments of the disclosure, by disposing the driving element in the non-exposed region in the transparent display device, the uniformity of the transmittance of the transparent display device in the exposed region may be improved.

[0007] In order to make the aforementioned features and advantages of the invention comprehensible, embodiments accompanied with drawings are described in detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0009] FIG. 1 is a schematic view of a transparent display device according to an embodiment of the disclosure.

[0010] FIG. 2 is a schematic view of a transparent display device in a first state according to an embodiment of the disclosure.

[0011] FIG. 3 is a schematic cross-sectional view of the transparent display device of FIG. 2 along the line III-III.

[0012] FIG. 4 is a schematic view of a transparent display device in a second state according to an embodiment of the disclosure.

[0013] FIG. 5 is a schematic cross-sectional view of the transparent display device of FIG. 4 along the line V-V.

[0014] FIG. 6 and FIG. 7 respectively are schematic views of an enlarged area E1 and an enlarged area E2 in FIG. 1 according to some embodiments.

[0015] FIG. 8 and FIG. 9 respectively are schematic views of the enlarged area E1 and the enlarged area E2 in FIG. 1 according to other embodiments.

[0016] FIG. 10 is a schematic view of part of wires in the transparent display device of FIG. 1.

[0017] FIG. 11 is a schematic view of a transparent display device according to another embodiment of the disclosure.

[0018] FIG. 12 is a schematic view of a transparent display device according to yet another embodiment of the disclosure.

[0019] FIG. 13 is a schematic view of an enlarged area E3 of FIG. 12 according to an embodiment.

[0020] FIG. 14 is a schematic cross-sectional view of the transparent display device of FIG. 12 along the line XIV-XIV according to some embodiments.

[0021] FIG. 15 is a schematic cross-sectional view of the transparent display device of FIG. 12 along the line XV-XV according to some embodiments.

[0022] FIG. 16 is a schematic view of a display panel according to some embodiments of the disclosure.

[0023] FIG. 17 is a schematic cross-sectional view of the display panel of FIG. 16 along the line XVII-XVII.

[0024] FIG. 18 is a schematic cross-sectional view of the display panel of FIG. 16 along the line XVIII-XVIII.

[0025] FIG. 19 is a schematic view of a transparent display device according to still another embodiment of the disclosure.

[0026] FIG. 20 is a schematic view of a transparent display device according to still yet another embodiment of the disclosure.

[0027] FIG. 21 is a schematic cross-sectional view of the transparent display device of FIG. 20 along the line XXI-XXI according to some embodiments.

DESCRIPTION OF THE EMBODIMENTS

[0028] Reference will now be made in detail to the exemplary embodiments of the disclosure, and examples of which are illustrated in the accompanying drawings. Whenever possible, the same reference numerals are used to represent the same or similar parts in the accompanying drawings and description.

[0029] Throughout the specification and the appended claims of the disclosure, certain terms are used to refer to specific elements. Those skilled in the art should understand that display device manufacturers may probably use different names to refer to the same elements. The specification is

not intended to distinguish between elements that have the same function but different names. In the following specification and claims, the terms “including” and “having”, etc., are open-ended terms, so they should be interpreted to mean “including but not limited to . . .”.

[0030] Directional wordings mentioned in the specification, such as “up,” “down,” “left,” “right,” “front,” and “back,” merely refer to directions in the accompanying drawings. Therefore, the directional wordings are used to illustrate rather than limit the disclosure. In the drawings, each drawing illustrates the general features of the methods, structures, and/or materials used in specific embodiments. However, the drawings should not be interpreted as defining or limiting the scope or nature covered by the embodiments. For example, for clarity, a relative size, a thickness, and a location of each film layer, region, and/or structure may be reduced or enlarged.

[0031] A structure (or layer, element, substrate) being located on another structure (or layer, element, substrate) described in the disclosure may mean that two structures are adjacent and directly connected, or may mean that two structures are adjacent and indirectly connected. Indirect connection means that there is at least one intermediate structure (or intermediate layer, intermediate element, intermediate substrate, intermediate spacing) between two structures, the lower surface of a structure is adjacent or directly connected to the upper surface of the intermediate structure, and the upper surface of the other structure is adjacent or directly connected to the lower surface of the intermediate structure. The intermediate structure may be a single-layer or multi-layer physical structure or non-physical structure, which is not limited. In the disclosure, when a structure is disposed “on” another structure, it may mean that a structure is “directly” disposed on another structure, or a structure is “indirectly” disposed on another structure, that is, at least one structure is sandwiched between a structure and another structure.

[0032] The terms “electrically connected to” or “coupled to” described in the disclosure may refer to direct connection or indirect connection. In the case of direct connection, the terminals of the elements on the two circuits are directly connected or connected to each other by a conductor line. In the case of indirect connection, there are switches, diodes, capacitors, inductors, resistors, other suitable elements, or a combination thereof between the terminals of the elements on the two circuits, but the disclosure is not limited thereto.

[0033] In the disclosure, the thickness, length, or width may be measured by an optical microscope, and the thickness or the width may be measured according to a cross-sectional image in an electron microscope, but the disclosure is not limited thereto. In addition, there may be a certain error between any two values or directions used for comparison. Moreover, in the disclosure, the terms such as “about”, “equal”, “same”, “substantially”, or “approximately” are generally interpreted as being within a range of plus or minus 15% of a given value or range, or as being within a range of plus or minus 5%, plus or minus 3%, plus or minus 2%, plus or minus 1%, or plus or minus 0.5% of the given value or range. In addition, the terms “the scope between the first value and the second value” and “ranging from the first value to the second value” mean that the range includes the first value, the second value, and other values in between.

[0034] In the disclosure, the features of multiple embodiments to be described below may be replaced, recombined, or mixed to form other embodiments without departing from the spirit of the disclosure. The features of multiple embodiments may be used in combination as long as such combination does not depart from the spirit of the disclosure or lead to conflict.

[0035] FIG. 1 is a schematic view of a transparent display device according to an embodiment of the disclosure. In FIG. 1, for example, a transparent display device **100** includes an exposed region **100A** and a non-exposed region **100B**. The non-exposed region **100B** may partially surround the exposed region **100A**. In some embodiments, the non-exposed region **100B** is substantially distributed around part of the periphery of the transparent display device **100** and may not surround the entire exposed region **100A**. In some embodiments, the exposed region **100A** may extend to part of the periphery of the transparent display device **100** without being completely surrounded by the non-exposed region **100B**. The non-exposed region **100B** may be interpreted as an area in the

transparent display device **100** that may be hidden by a frame (not shown in FIG. 1) or may be disposed inside the frame. In some embodiments, in an actual implementation, the non-exposed region **100B** may be hidden and may not be directly seen by the user. The exposed region **100A** may be interpreted as an area in the transparent display device **100** that may be exposed in an actual implementation. However, the disclosure does not exclude the situation when the exposed region **100A** is temporarily hidden as the transparent display device is in operation. For example, in an actual implementation, the transparent display device **100** may have different operation states. In some operation states, at least part of the exposed region **100A** may be exposed for the user to see it directly. In other operation states, the exposed region **100A** that can be seen by the user may be partially or completely hidden. In other words, when the transparent display device **100** is in operation, the exposed region **100A** may be hidden or exposed, but the non-exposed region **100B** may be hidden in any operation state. For ease of comprehension, in FIG. 1 and the subsequent drawings, the orientations of the transparent display device **100** in the respective drawings are illustrated in the X direction, the Y direction, and the Z direction. The Y direction may be perpendicular to the upper surface or the lower surface of the transparent display device **100**, and the X direction and the Z direction may be parallel to the upper surface or the lower surface of the transparent display device **100**. The Y direction may be perpendicular to the X direction and the Z direction, and the X direction may be perpendicular to the Z direction. In the embodiment, the plane of the transparent display device **100** is illustrated as a plane oriented in the X direction and the Z direction.

[0036] The transparent display device **100** may at least include a display panel **120** and a driving element DR. The display panel **120** may include multiple pixels PX. For example, the pixels PX are disposed in the exposed region **100A**, and the driving element DR is disposed in the non-exposed region **100B**. The driving element DR is adapted for driving the pixels PX. The pixels PX may be adopted to emit light to display images, and the driving element DR may be adopted to transmit the signals required by the pixels PX to the pixel. In some embodiments, for example, the pixels PX may include liquid crystals, organic light-emitting diodes (OLEDs), inorganic light-emitting diodes (LEDs), mini-LEDs, micro-LEDs, quantum dots (QDs), quantum dot diodes (QLEDs/QDLEDs), electro-phoretic, fluorescence, phosphors, other suitable materials, or a combination thereof, but the disclosure is not limited thereto. In some embodiments, the multiple pixels PX may emit light of multiple colors to achieve an effect of colorful display. The driving element DR may be an opaque element, so disposing the driving element DR in the non-exposed region **100B** may reduce the range of the opaque area defined by the driving element DR in the exposed region **100A** of the transparent display device **100**, which contributes to improving the overall transmittance of the exposed region **100A**, and/or increasing the area of the exposed region **100A** that may exhibit a transparent effect. The transparent display device in the disclosure may be applied to various fields, such as buildings, automobiles, interior decoration, signboards, shop windows, or optical devices, but the disclosure is not limited thereto.

[0037] In FIG. 1, the display panel **120** of the transparent display device **100** may be disposed on a carrier **110**. For example, the carrier **110** may be a plate with translucency and sufficient supporting properties. In some embodiments, the material of the carrier **110** may include glass, quartz, sapphire, polymer (e.g., polyimide (PI)), polyethylene terephthalate (PET) and/or other suitable materials, a combination thereof, or the like, and the disclosure is not limited thereto. In some embodiments, the carrier **110** may have a single-layer or multi-layer structure. Specifically, the display panel **120** may be attached to or fabricated on the carrier **110**. In some embodiments, the display panel **120** may be a transparent display panel. In other words, the display panel **120** has a certain transmittance so that the user may see the view behind the display panel **120**. In some embodiments, the area of the display panel **120** may be less than or equal to the area of the carrier **110**, but the disclosure is not limited thereto.

[0038] The display panel **120** may include a display area **120A** and a non-display area **120B**. The

pixels PX are disposed in the display area **120A** to display images in the display area **120A**, and the non-display area **120B** may surround the display area **120A**. The display area **120A** overlaps the exposed region **100A** of the transparent display device **100**, at least part of the non-display area **120B** overlaps the exposed region **100A**, and another part overlaps the non-exposed region **100B**. The non-display area **120B** may include a bonding region BR disposed in the non-exposed region **100B**. Moreover, in addition to the pixels PX disposed on the display panel **120**, data connection lines SC1 and scan connection lines SC2 are further disposed thereon. Specifically, signal lines (not shown) corresponding to the pixels PX, such as scan lines and data lines, are disposed in the display area **120A**, and the data connection lines SC1 and the scan connection lines SC2 may be connected to the signal lines disposed in the display area **120A**. The data connection lines SC1 and the scan connection lines SC2 may extend outward from the periphery of the display area **120A**, and at least a part of the lines extend to the bonding region BR.

[0039] The driving element DR may include a data driving element, a gate driving element, a driving carrier board DRB, and a connecting element DRC. The data driving element may include a driving circuit DR1; and the gate driving element may include a driving circuit DR2, the driving carrier board DRB, and the connecting element DRC. The driving circuit DR1 may be disposed on the driving carrier board DRB. The driving circuit DR1 may include an integrated circuit element, but the disclosure is not limited thereto. Moreover, the driving carrier board DRB may be a circuit board, such as a printed circuit board, but the disclosure is not limited thereto. The driving carrier board DRB may be bonded to the bonding region BR of the display panel **120** through one or more connecting elements DRC and connected to the data connection lines SC1. For example, the connecting element DRC may include a flexible circuit board, but the disclosure is not limited thereto. Therefore, the driving circuit DR1 of the driving element DR may transmit the corresponding signals to the pixels PX through the connecting element DRC and the multiple data connection lines SC1. In addition, the driving circuit DR2 may be disposed on the display panel **120**, and the driving circuit DR2 of the driving element DR may transmit multiple signals to the pixels PX through multiple scan connection lines SC2.

[0040] In some embodiments, the driving circuit DR1 may include a data signal driving circuit for providing the data signal required by the pixel PX, and the driving circuit DR2 may include a scan signal driving circuit for providing the scan signal required by the pixel PX. In some embodiments, the driving circuit DR1 may be a packaged integrated circuit element, and the driving circuit DR2 may include elements, such as multiple transistors, multiple capacitors, and other elements fabricated on the display panel **120**, but the disclosure is not limited thereto. In some embodiments, the driving circuit DR2 does not have an independent package structure but is integrated in the circuit layer of the pixel PX, but it is not limited thereto. In other embodiments, for example, the driving circuit DR2 (the scan signal driving circuit) may be implemented as a packaged integrated circuit element like the driving circuit DR1, or it may be integrated in the packaged integrated circuit element of the driving circuit DR1.

[0041] In the embodiment, the driving elements DR are all disposed in the non-exposed region **100B** of the transparent display device **100**, which contributes to improving the transmittance of the exposed region **100A**, such as the transmittance of visible light, or to improving the uniformity of the transmittance of the exposed region **100A**. Therefore, when in operation, the transparent display device **100** may have good performance in light transmission, and since the transmittance of the exposed region **100A** is uniform, the user may see the environment behind the transparent display device **100** clearly. For example, in a rectangular range RR having the largest area in the range of the exposed region **100A**, the center point RR1 of the rectangular range RR may be defined; along the Z direction, the center point RR1 is projected to the edge points RR2 and RR3 of the edge of the rectangular range RR; and along the X direction, the center point RR1 is projected to the edge points RR4 and RR5 of the edge of the rectangular range RR. The difference in the transmittance of the transparent display device **100** at the center point RR1, the edge point RR2, the edge point

RR3, the edge point RR4, and the edge point RR5 may be within 30%. For example, $|(transmittance\ TRR_i - transmittance\ TRR_j)| / transmittance\ TRR_i * 100\% \leq 30\%$, where i and j are any two of 1, 2, 3, 4, and 5. In the embodiment, the “transmittance” refers to the percentage of which the light intensity of transmitted light measured after the ambient light penetrates the transparent display device **100** is divided by the light intensity of the measured ambient light that does not penetrate the transparent display device **100**. The “light intensity” refers to the spectrum integral value of the light source (e.g., display light or ambient light). In some embodiments, the light source may include visible light (e.g., light with a wavelength ranging from 380 nm to 780 nm) or ultraviolet light (e.g., light with a wavelength less than 365 nm), but the disclosure is not limited thereto. That is, when the light source is visible light, the light intensity is the spectrum integral value with a wavelength ranging from 380 nm to 780 nm. In other embodiments, when two area ranges of the same area are arbitrarily selected from the exposed region **100A** of the transparent display device **100**, the transmittance of the two area ranges is approximately similar or the same. For example, when selecting an area range of a specific area size in the display area **120A** of the exposed region **100A** and an area range of the same specific area size in the non-display area **120B** of the exposed region **100A**, the transmittance of the two areas may be approximately the same or may differ by less than 30%.

[0042] According to some embodiments, with the design in which the transparent display device **100** is disposed on the carrier **110**, the transmittance relation of the center point RR1, the edge point RR2, the edge point RR3, the edge point RR4, and the edge point RR5 may also conform to the relation: $|(transmittance\ TRR_i - transmittance\ TRR_j)| / transmittance\ TRR_i * 100\% \leq 30\%$, where i and j are any two of 1, 2, 3, 4, and 5. In other words, whether the transparent display device **100** is disposed on the carrier **110** or not, the transmittance corresponding to the center point RR1, the edge point RR2, the edge point RR3, the edge point RR4, and the edge point RR5 are approximately the same; and the effect of uniform transmittance may be achieved. That is, the transmittance of the carrier **110** at each position is approximately the same. FIG. 2 is a schematic view of a transparent display device in a first state according to an embodiment of the disclosure; and FIG. 3 is a schematic cross-sectional view of the transparent display device of FIG. 2 along the line III-III. In FIG. 2, an electronic device ED may include the transparent display device **100**, the carrier **110**, and a frame **200**. The transparent display device **100** may be disposed in the frame **200**. In the first state of FIG. 2, the transparent display device **100** may be accommodated in the frame **200**. Meanwhile, the transparent display device **100** may be completely hidden by the frame **200**. The elements of the transparent display device **100** in FIG. 2 may be the same as or similar to those in FIG. 1, and they are not iterated herein. In FIG. 3, the transparent display device **100** may include the display panel **120** and the driving element DR. The display panel **120** may include a substrate **122**, an exposed region element **124**, and a non-exposed region element **126**. In some embodiments, the exposed region element **124** may include the pixels PX shown in FIG. 1 and related signal lines connected to the pixels PX. In some embodiments, the exposed region element **124** may include a display element, and may also include a touch element, a sensing element, and the like. The non-exposed region element **126** may include the data connection lines SC1 and the scan connection lines SC2 shown in FIG. 1. The driving element DR may include the driving circuit DR1, the driving circuit DR2, the driving carrier board DRB, and the connecting element DRC shown in FIG. 1, the driving circuit DR2 shown in FIG. 1 may also be disposed in the display panel **120** as a part of the exposed region element **124**. In addition, the display panel **120** of the transparent display device **100** may further include a protection layer **128** disposed on the substrate **122** and covering the exposed region element **124** and the non-exposed region element **126** to reduce the probability of damage to the exposed region element **124** and the non-exposed region element **126**.

[0043] The substrate **122** may be a multi-layer substrate including multiple layer structures. In some embodiments, the layer structures of the substrate **122** may include an inflexible substrate, a

flexible substrate, an insulating layer, and a conductive layer, or any combination thereof. The substrate **122** may be a rigid substrate, a flexible substrate, or a combination thereof. Moreover, for example, the material of the substrate **122** may include glass, quartz, ceramic, sapphire, plastic, polycarbonate (PC), polyimide (PI), polypropylene (PP), polyethylene terephthalate (PET), other suitable materials, or a combination thereof, but the disclosure is not limited thereto. In some embodiments, at least one of the layer structures of the substrate **122** may have multiple slits or holes, and the slits or holes are disposed in the exposed region **100A** of the transparent display device **100**. In other embodiments, at least one of the layer structures of the substrate **122** may have slits or holes in the non-exposed region **100B**. The disposition of slits or holes in the substrate **122** contributes to improving the flexibility, transparency, and/or stretchability of the substrate **122**, so that the display panel **120** may conform to the surface of structures with different curvatures or conform to the surface of different structures in irregular shapes. In some embodiments, the disposition of slits or holes in the substrate **122** may also contribute to improving the transmittance of the display panel **120**. In addition, the protection layer **128** may adopt different materials in different regions. For example, the protection layer **128** may include a light-transmitting material in the exposed region **100A** and may include an opaque/shielding material in the non-exposed region **100B**. For example, the transmittance of the opaque/shielding material is less than that of the light-transmitting material.

[0044] According to FIG. 3, the frame **200** may accommodate the transparent display device **100**, and the frame **200** has an opening **202**. In some embodiments, a driving mechanism (not shown) may be disposed in the frame **200**, and the transparent display device **100** may be disposed on the driving mechanism, so that the driving mechanism may push the transparent display device **100** to move in the direction Z, and therefore, the transparent display device **100** may protrude from the opening **202** of the frame **200** to expose the exposed region **100A** or the transparent display device **100** is completely hidden in the frame **200**. In some embodiments, the design of disposing the transparent display device **100** in the frame **200** may be applied to windows, such as car windows, but the disclosure is not limited thereto. In other embodiments, the design of disposing the transparent display device **100** in the frame **200** may be applied to display windows or similar products.

[0045] FIG. 4 is a schematic view of a transparent display device in a second state according to an embodiment of the disclosure; and FIG. 5 is a schematic cross-sectional view of the transparent display device of FIG. 4 along the line V-V. The elements shown in FIG. 4 are the same as those in FIG. 2, and the elements shown in FIG. 5 are the same as those in FIG. 3. However, it is illustrated that the transparent display device **100** is in the second state in FIG. 4 and FIG. 5. In the second state, the transparent display device **100** may be moved and then exposed outside the frame **200**. In some embodiments, the transparent display device **100** may be moved by the driving mechanism disposed in the frame **200** to be in the state in FIG. 4 and FIG. 5. In the state in FIG. 4 and FIG. 5, the area of the transparent display device **100** not hidden by the frame **200** is the exposed region **100A** shown in FIG. 1, and the area of the transparent display device **100** hidden by the frame **200** is the non-exposed region **100B** shown in FIG. 1. For example, the state shown in FIG. 4 and FIG. 5 is a state when the driving mechanism is at the limit of movement. That is, the driving mechanism may no longer move the transparent display device **100** further away from the frame **200** in the Z direction from the state of FIG. 4 and FIG. 5. For example, the exposed region **100A** is defined in the state of FIG. 4 and FIG. 5, and the boundary of the exposed region **100A** may be defined along the boundary of the frame **200** in the state of FIG. 4 and FIG. 5.

[0046] In addition, in another state, the relative positions of the transparent display device **100** and the frame **200** may be between those positions shown in FIG. 2 and FIG. 4. Meanwhile, part of the area of the exposed region **100A** may be hidden by the frame **200**, and the non-exposed region **100B** is hidden by the frame **200** in any state. Therefore, when the user is using the device, the non-exposed region **200** may not be seen by the user, and part of the exposed region **100** or the entire

exposed region **100** may be seen by the user according to the switch of the operation state. In some embodiments, the transparent display device **100** may be applied to a car window, the first state shown in FIG. **2** and FIG. **3** is a state when the car window is fully opened, and the second state in FIG. **4** and FIG. **5** is a state when the car window is fully closed. According to the description, the exposed region **100A** has uniform transmittance, so when the transparent display device **100** is applied to a car window, the car window may have uniform transmittance to display a desired visual effect.

[0047] FIG. **6** and FIG. **7** respectively are schematic views of an enlarged area **E1** and an enlarged area **E2** in FIG. **1** according to some embodiments. In FIG. **6**, three pixels **PX**, including a pixel **PXR**, a pixel **PXG**, and a pixel **PXB**, respectively, may be disposed in the enlarged area **E1**. In addition, the enlarged area **E1** also includes multiple first signal lines **SL1** and multiple second signal lines **SL2**. The pixel **PXR**, the pixel **PXG**, and the pixel **PXB** are each a light-emitting pixel capable of emitting light for displaying images. In some embodiments, the pixel **PXR**, the pixel **PXG**, and the pixel **PXB** may emit light of different colors, such as red light, green light, blue light, etc., but the disclosure is not limited thereto. For example, each of the first signal lines **SL1** extends in the **Z** direction, and each of the second signal lines **SL2** extends in the **X** direction, for example. The pixel **PXR**, the pixel **PXG**, and the pixel **PXB** may share one of the second signal lines **SL2**, and respectively correspond to different first signal lines **SL1**. Specifically, FIG. **6** illustrates a layout in which a sequence of the first signal line **SL1**, the pixel **PXR**, the first signal line **SL1**, the pixel **PXG**, the first signal line **SL1**, and the pixel **PXB** in the order is arranged along the **X** direction, but it is not limited thereto.

[0048] In the embodiment, the enlarged area **E1** further includes a pixel transparent area **TPX**. The pixel transparent area **TPX** refers to an area range without signal lines and pixels **PX**. That is, the user may see through the transparent display device **100** in the pixel transparent area **TPX**. In FIG. **6**, the pixels **PX** are disposed in a centralized manner, so the pixel transparent area **TPX** is disposed on the same side of the pixel **PXR**, the pixel **PXG**, and the pixel **PXB**, but it is not limited thereto. In some embodiments, a transmittance adjustment layer (not shown) may be further disposed in the pixel transparent area **TPX** or an area corresponding to the pixel transparent area **TPX** (e.g., an area overlapping the pixel transparent area **TPX** when viewed along the direction **Y**). The transmittance adjustment layer may be disposed between the substrate **122** and the carrier **110** in the cross-sectional structure of FIG. **3**. The transmittance adjustment layer may control its transmittance through electrical signals. For example, the material of the transmittance adjustment layer may include dichroic dye liquid crystals (DDLCs), polymer dispersed liquid crystals (PDLCs), polymer network liquid crystals (PNLCs), cholesteric liquid crystals (CLCs), electrochromic (EC) materials, suspended particle devices (SPDs), or a combination thereof.

[0049] The transmittance adjustment layer may improve the visible contrast of the transparent display device **100**. For example, in an environment where the light intensity of the ambient light is high, the transmittance of the transmittance adjustment layer may be reduced, so the ambient light is shielded and it is easier for the image displayed on the transparent display device **100** to be recognized. In addition, in some embodiments, when the transparent display device **100** is applied to products such as car windows or windows and when the light intensity of the ambient light is too high, the transmittance of the transmittance adjustment layer may be reduced to shield the ambient light to improve the visual comfort of passengers. Alternatively, reducing the transmittance of the transmittance adjustment layer contributes to improving the privacy of the passengers, but the disclosure is not limited thereto.

[0050] The enlarged area **E2** is mainly disposed in the non-display area **120B** shown in FIG. **1**, multiple scan connection lines **SC2** may be disposed in the enlarged area **E2**, and each of the scan connection lines **SC2** may be connected to one of the second signal lines **SL2**. According to the layout of FIG. **1**, each of the scan connection lines **SC2** may be adapted to electrically connect one of the second signal lines **SL2** to the driving circuit **DR2**. In addition, each of the scan connection

lines SC2 may be divided into an exposed section SC2A of the scan connection line and a hidden section SC2B of the scan connection line. The exposed section SC2A of the scan connection line refers to the section of the scan connection line SC2 disposed in the exposed region **100A**, and the hidden section SC2B of the scan connection line refers to the section of the scan connection line SC2 disposed in the non-exposed region **100B**. In the enlarged area E2, each exposed section SC2A of the scan connection line of the scan connection line SC2 is illustrated. The layout of the exposed section SC2A of the scan connection line in the enlarged area E2 is being disposed in groups, for example. For example, several exposed sections of the scan connection line SC2A may be disposed in a centralized manner to form a group GSC of scan connection lines, a distance SSC separates adjacent groups GSC of scan connection lines, and there is no connection line in the distance SSC. Therefore, the distance SSC may define a connection line transparent area TSC.

[0051] In some embodiments, the enlarged area E1 of FIG. 6 and the enlarged area E2 of FIG. 7 may have the same area, and the pixel transparent area TPX in the enlarged area E1 and the connection line transparent area TSC in the enlarged area E2 may have an approximately same or same area. In this way, the transmittance of the enlarged area E1 and the transmittance of the enlarged area E2 may be approximately the same, so the design of the uniform transmittance of the exposed region **100A** (as shown in FIG. 1) may be achieved.

[0052] FIG. 8 and FIG. 9 respectively are schematic views of the enlarged area E1 and the enlarged area E2 in FIG. 1 according to other embodiments. The elements shown in FIG. 8 and FIG. 9 are the same as those in FIG. 6 and FIG. 7. However, the layout of the elements in FIG. 8 and FIG. 9 is different from the layout of the elements in FIG. 6 and FIG. 7. In FIG. 8, the pixels PXR, the pixels PXG, and the pixels PXB are disposed in the enlarged area E1 in a substantially equally spaced manner; and the pixel transparent area TPX may be divided into the pixel transparent area TPXR beside the pixels PXR, the pixel transparent area TPXG beside the pixels PXG, and the pixel transparent area TPXB beside the pixels PXB. In FIG. 9, the exposed sections SC2A of the scan connections line are disposed in the enlarged area E2 in a substantially equally spaced manner, and the connection line transparent area TSC is divided into multiple regions by the exposed sections SC2A of the scan connection lines. Overall, when the enlarged area E1 and the enlarged area E2 have similar areas, the overall area of the connection line transparent area TSC may be substantially similar or even equal to the overall area of the pixel transparent area TPX, so the entire exposed region **100A** (shown in FIG. 1) has uniform transmittance.

[0053] FIG. 10 is a schematic view of part of wires in the transparent display device of FIG. 1. Specifically, FIG. 10 illustrates a schematic view of a signal line SL disposed in the display area **120A** and a signal connection line SC disposed in the non-display area **120B** in the transparent display device **100** of FIG. 1. In some embodiments, the signal line SL may be interpreted as an implementation of any first signal line or any second signal line in the enlarged area E1, and the signal connection line SC may be interpreted as an implementation of any scan connection line SC2 in the enlarged area E2, but the disclosure is not limited thereto. In some embodiments, the signal line SL and the signal connection line SC may be different sections of the same conductor line. In FIG. 10, the signal line SL may be a wire with a solid pattern, and the signal connection line SC may be a wire with multiple holes VSC, which accordingly contributes to improving the overall transmittance of the non-display area **120B**. In some embodiments, a line width WSC of the signal connection line SC may be greater than a line width WSL of the signal line SL, but the disclosure is not limited thereto. Meanwhile, the measurement of the line width may be interpreted that when a section of the wire extends along an extension direction, the maximum width of the section of the wire in the vertical direction of the extension direction is the line width.

[0054] FIG. 11 is a schematic view of a transparent display device according to another embodiment of the disclosure. In FIG. 11, the elements of a transparent display device **102** are substantially the same as those of the transparent display device **100**, so the same element reference numerals in the two embodiments are interpreted as the same elements. Specifically, the transparent

display device **102** includes the display panel **120** and the driving element DR disposed on the carrier **110**. The driving element DR may include a data driving element, a gate driving element, a driving carrier board DRB, and a connecting element DRC. The data driving element may include a driving circuit DR1, and the gate driving element may include a driving circuit DR2. The difference between the transparent display device **102** and the transparent display device **100** is where the driving circuit DR2 is disposed. In FIG. **11**, the driving circuit DR2 may be disposed in the exposed region **100A**, and specifically disposed in the non-display area **120B** of the display panel **120**. The driving circuit DR2 may be electrically connected to the signal line disposed in the display area **120A** of the display panel **120**. In some embodiments, the driving circuit DR1 may be connected to the driving circuit DR2 through a corresponding signal connection line SC3.

[0055] FIG. **12** is a schematic view of a transparent display device according to yet another embodiment of the disclosure. In FIG. **12**, the elements of a transparent display device **104** are substantially the same as those of the transparent display device **100**, so the same element reference numerals in the two embodiments are interpreted as the same elements. Specifically, the transparent display device **102** includes the display panel **120** and the driving element DR disposed on the carrier **110**. The driving element DR may include a data driving element, a gate driving element, a driving carrier board DRB, and a connecting element DRC. The data driving element may include a driving circuit DR1, and the gate driving element may include a driving circuit DR2. The difference between the transparent display device **104** and the transparent display device **100** is where the driving circuit DR2 and the multiple scan connection lines SC2 are disposed. In FIG. **12**, the driving circuit DR2 and the multiple scan connection lines SC2 may be disposed in the non-exposed region **100B**, and the driving circuit DR2 is adopted to be electrically connected to the signal line disposed in the display area **120A** of the display panel **120**.

[0056] FIG. **13** is a schematic view of an enlarged area E3 of FIG. **12** according to an embodiment. According to FIG. **12** and FIG. **13**, multiple pixels PX, multiple first signal lines SL1, and multiple second signal lines SL2 are disposed in the exposed region **100A** in the transparent display device **104**. The multiple pixels PX are disposed along the X direction, for example, and a first signal line SL1 and a second signal line SL2 may be disposed between two adjacent pixels PX. The first signal line SL1 and the second signal line SL2 are adopted to transmit different signals, but they generally extend along the Z direction. In some embodiments, the first signal line SL1 may be adopted to transmit data signals, and the second signal line SL2 may be adopted to transmit scan signals. In the embodiment of FIG. **12** and FIG. **13**, the first signal line SL1 and the second signal line SL2 both extend in the same direction and extend toward the non-exposed region **100B**. The driving circuit DR2 is not disposed in the exposed region **100A**, which contributes to improving the uniformity of the transmittance of the exposed region **100A**.

[0057] According to FIG. **12** and FIG. **13**, the data connection line SC1 disposed in the non-exposed region **100B** is connected to the first signal line SL1 in the exposed region **100A**, for example, and the scan connection line SC2 disposed in the non-exposed region **100B** is connected to the second signal line in the exposed region **100A**, for example. In addition, the driving circuit DR2 may be disposed in the non-exposed region **100B** and between the exposed region **100A** and the bonding region BR. The scan connection line SC2 extends between the exposed region **100A** and the driving circuit DR2 to connect the second signal line SL2 to the driving circuit DR2. The data connection line SC1 extends between the first signal line SL1 of the exposed region **100A** and the bonding region BR. Therefore, the data connection line SC1 and the scan connection line SC2 may partially or completely overlap in the Y direction.

[0058] FIG. **14** is a schematic cross-sectional view of the transparent display device of FIG. **12** along the line XIV-XIV according to some embodiments; and FIG. **15** is a schematic cross-sectional view of the transparent display device of FIG. **12** along the line XV-XV according to some embodiments. As shown in FIG. **14** and FIG. **15**, the transparent display device **104** may be disposed on the display panel **120** and the driving circuit DR2 on the carrier **110**. The display panel

120 includes the substrate **122**, the data connection line **SC1**, the scan connection line **SC2**, an insulating layer **IN1**, and an insulating layer **IN2**. The scan connection line **SC2** and the driving circuit **DR2** are both disposed on the substrate **122**, the insulating layer **IN1** covers the scan connection line **SC2** and the driving circuit **DR2**, the data connection line **SC1** is disposed on the insulating layer **IN1**, and the insulating layer **IN2** covers the data connection line **SC1**. In this way, the insulating layer **IN1** may separate the scan connection line **SC2** from the data connection line **SC1**, and also separate the driving circuit **DR2** from the data connection line **SC1**. The insulating layer **IN1** and the insulating layer **IN2** may include a single-layer or multi-layer structure and for example, may include organic materials, inorganic materials, or a combination thereof, but the disclosure is not limited thereto. The organic materials may include polyethylene terephthalate (PET), polyethylene (PE), polyethersulfone (PES), polycarbonate (PC), polymethylmethacrylate (PMMA), polyimide (PI), photo sensitive polyimide (PSPI) or a combination thereof; and the inorganic material may include silicon nitride, silicon oxide, silicon oxynitride, or a combination thereof, but the disclosure is not limited thereto.

[0059] FIG. **16** is a schematic view of a display panel according to some embodiments of the disclosure. A display panel **120'** in FIG. **16** is an embodiment of the display panel **120** in FIG. **1**, for example, so it may be applied to the transparent display device **100** in FIG. **1**. The display panel **120'** includes a display area **120A'** and a non-display area **120B'**, the layout of the display area **120A'** and the non-display area **120B'** is substantially similar to that of the display area **120A** and the non-display area **120B** of FIG. **1**, and the pixels **PX** are disposed in the display area **120A'** of the display panel **120'**. The display panel **120'** includes a substrate **122'**, for example, and the substrate **122'** is a mesh substrate. In some embodiments, the substrate **122'** may be a flexible substrate, but it is limited thereto. When the display panel **120** is applied to the transparent display device **100** of FIG. **1**, and the carrier **110** of FIG. **1** is a carrier **110** with a curved surface, the structure and flexibility of the substrate **122'** contribute to enabling the display panel **120'** to conform to and attach to the surface of the carrier **110** without causing undesired warpage or bending. In some embodiments, the substrate **122'** may have stretchable properties.

[0060] According to FIG. **16**, the substrate **122'** of the display panel **120'** may include multiple island-shaped portions **ISP** and a connection portion **CTP** connected between the island-shaped portions **ISP**. The pixels **PX** may be disposed on the island-shaped portions, and multiple pixels **PX** may be disposed on each of the island-shaped portions **ISP**. There may be no pixel **PX** in the connection portion **CTP**, but the disclosure is not limited thereto. A signal line may be disposed in the connection portion **CTP**, and the signal line may be adopted to transmit the signal required by the pixel **PX**. The substrate **122'** has a stretchable property, and the island-shaped portions **ISP** may be rotated in a state when the substrate **122'** is stretched. Meanwhile, the connection portion **CTP** may correspondingly be deformed due to the stretch, but the disclosure is not limited thereto. In some embodiments, the display panel **120'** may have a greater thickness at the island-shaped portion **ISP** where the pixel **PX** is disposed, and may have a lesser thickness at the connection portion **CTP**.

[0061] FIG. **17** is a schematic cross-sectional view of the display panel of FIG. **16** along the line XVII-XVII; and FIG. **18** is a schematic cross-sectional view of the display panel of FIG. **16** along the line XVIII-XVIII. According to FIG. **17** and FIG. **18**, the substrate **122'** may include a first flexible substrate **SB1** and a second flexible substrate **SB2**. However, in other embodiments, the substrate **122'** may include a single-layer flexible substrate. In addition, the pixel **PX** may include an active element **TFT**, a light-emitting element **LE**, and a connection electrode **CE** disposed on the island-shaped portion **ISP** of the substrate **122'**. The active element **TFT** includes a semiconductor layer **SE**, a gate electrode **GE**, a source electrode **SR**, and a drain electrode **DE**; and the light-emitting element **LE** includes an anode **AN**, a light-emitting layer **EL**, and a cathode **CT**.

[0062] The semiconductor layer **SE** and the gate electrode **GE** overlap in the **Y** direction and are separated from each other through an insulating layer **IN3**. The gate electrode **GE** is covered by an

insulating layer IN4, and the source electrode SR and the drain electrode DE are both disposed on the insulating layer IN4. The insulating layer IN3 and the insulating layer IN4 may be penetrated by vias V1 and V2 so that the source electrode SR and the drain electrode DE contact different parts of the semiconductor layer SE. An insulating layer IN5 covers the source electrode SR and the drain electrode DE.

[0063] The anode AN is disposed on the insulating layer IN5, and the insulating layer IN5 may be penetrated by a via V3 so that the anode AN may contact the drain electrode DE. A pixel definition layer PDL is also disposed on the insulating layer IN5, and at least part of the area of the anode AN is not covered by the pixel definition layer PDL. The light-emitting layer EL is disposed on the anode AN and surrounded by the pixel definition layer PDL. The cathode CT covers the light-emitting layer EL and the pixel definition layer PDL.

[0064] In addition, a signal line SL' may be disposed between the first flexible substrate SB1 and the second flexible substrate SB2. The signal line SL' may continuously extend between the adjacent island-shaped portions ISP in the display area 120A' and pass through the connection portion CTP. An insulating layer IN6 is further disposed between the second flexible substrate SB2 and the semiconductor layer SE. The insulating layer IN3, the insulating layer IN6, and the second flexible substrate SB2 may be penetrated through a via V4 so that the connection electrode CE contacts the signal line SL'. In some embodiments, the connection electrode CE may be connected to the gate electrode GE, and the signal line SL' is adopted to transmit scan signals. Alternatively, the connection electrode CE may be connected to the source electrode SR, and the signal line SL' is adopted to transmit data signals. In FIG. 18, the signal connection line SC' may be a conductor line disposed in the non-display area 120B' and electrically connected to the signal line SL'. The signal connection line SC' and the signal line SL' may include the same conductor layer.

[0065] FIG. 19 is a schematic view of a transparent display device according to still another embodiment of the disclosure. In FIG. 19, a transparent display device 106 is substantially the same as the transparent display device in FIG. 11, so the same element reference numerals in the two embodiments are interpreted as the same elements. Specifically, the difference between the transparent display device 106 of FIG. 19 and the transparent display device 102 of FIG. 11 is that a boundary B100 between the exposed region 100A and the non-exposed region 100B of the transparent display device 106 is non-linear. In addition, the driving element DR of the transparent display device 106 may include an additional driving carrier board DRD, which is bonded to the display panel 120 of the transparent display device 106 through a connecting element DRE. Specifically, the driving carrier board DRB and the driving carrier board DRD may be disposed on different sides of the carrier 110. Since the driving carrier board DRB and the driving carrier board DRD are both bonded and disposed in the non-exposed region 100B, the transparent display device 106 may have a relatively uniform transmittance in the exposed region 100A. In some embodiments, the transparent display device 106 may be disposed in the frame (not shown), and the boundary B100 is defined according to the outline of the frame, for example.

[0066] FIG. 20 is a schematic view of a transparent display device according to still yet another embodiment of the disclosure. In FIG. 20, a transparent display device 108 is substantially the same as the transparent display device 100 in FIG. 1, so the same element reference numerals in the two embodiments are interpreted as the same elements, and they are not iterated herein. In addition to all the elements of the transparent display device 100 (the display panel 120 and the driving element DR), the transparent display device 108 further includes a slit blocking structure 130, and the slit blocking structure 130 may be disposed along the periphery of the display panel 120.

[0067] FIG. 21 is a schematic cross-sectional view of the transparent display device of FIG. 20 along the line XXI-XXI according to some embodiments. The transparent display device 108 includes the display panel 120. The display panel 120 may be disposed on the carrier 110 and includes the substrate 122, an insulating layer stack INX, the pixel PX, and a protection layer PR. The insulating layer stack INX is disposed on the substrate 122, the pixel PX is disposed on the

insulating layer stack INX, and the protection layer PR is disposed on the pixel PX to cover the pixel PX. In the embodiment, the substrate **122** may be a multi-layer substrate including multiple layer structures. In some embodiments, the layer structures of the substrate **122** may include a supporting board, a flexible substrate, an insulating layer, and/or a conductive layer. In some embodiments, at least one of the layer structures of the substrate **122** may have multiple slits or holes, and the slits or holes are disposed in the exposed region **100A** of the transparent display device **108**. The insulating layer stack INX may include a stack of multiple insulating layers. In some embodiments, at least one insulating layer in the insulating layer stack INX and the insulating layer of the substrate **122** may have the same material. The protection layer PR may adopt different materials in different regions. For example, the protection layer PR may include a light-transmitting material in the exposed region **100A** and may include an opaque/shielding material in the non-exposed region **100B**. For example, the transmittance of the opaque/shielding material is less than that of the light-transmitting material.

[0068] The structure of the pixel PX may refer to the related description of FIG. **17**, but the disclosure is not limited thereto. The protection layer PR covers all elements on the substrate **122**. Specifically, the pixels PX are disposed in the display area **120A** to display images in the display area **120A**, and the non-display area **120B** may surround the display area **120A**. The pixel PX may include the active element TFT and the light-emitting element LE. The active element TFT includes a semiconductor layer SE, a gate electrode GE, a source electrode SR, and a drain electrode DE; and the light-emitting element LE includes an anode AN, a light-emitting layer EL, and a cathode CT.

[0069] In addition, the transparent display device **108** further includes the slit blocking structure **130** disposed in the insulating layer stack INX. For example, the slit blocking structure **130** is a groove-shaped structure formed in the insulating layer stack INX. The slit blocking structure **130** may penetrate all or part of the insulating layers of the insulating layer stack INX. The number of the slit blocking structure **130** may be more than one, but may also be one. The protection layer PR may cover the slit blocking structure **130** and the insulating layer stack INX defining the slit blocking structure **130**.

[0070] Based on the above, the transparent display device of the embodiments in the disclosure includes an exposed region and a non-exposed region. The non-exposed region refers to an area that is hidden by the frame and may not be seen by the user when the device is in operation. In the transparent display device of the embodiments in the disclosure, at least part of the driving elements, such as driving circuits may be disposed in the non-exposed region. In this way, the exposed region of the transparent display device does not include a large-sized shielding element, which contributes to improving the uniformity of the transmittance of the exposed region. In addition, the exposed region of the transparent display device may also provide good light transmittance.

[0071] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims and their equivalents.

Claims

1. An electronic device, comprising: a display panel having a display area and a non-display area adjacent to the display area, wherein the display panel comprises: a pixel disposed in the display area; a data connection line disposed in the non-display area; and a scan connection line disposed in the non-display area; a data signal driving circuit, wherein the data connection line is electrically connected between the pixel and the data signal driving circuit; and a scan signal driving circuit,

wherein the scan connection line is electrically connected between the pixel and the scan signal driving circuit, wherein the data connection line and the scan connection line in the non-display area are at least partially overlapped.

2. The electronic device according to claim 1, wherein the data signal driving circuit comprises a packaged integrated circuit element.

3. The electronic device according to claim 1, wherein the scan signal driving circuit comprises a packaged integrated circuit element.

4. The electronic device according to claim 1, further comprising a driving carrier board bonded to the display panel, the data signal driving circuit is disposed on the driving carrier board.

5. The electronic device according to claim 1, further comprising a first insulating layer, located between the scan connection line and the data connection line.

6. The electronic device according to claim 5, further comprising a second insulating layer located on the first insulating layer and the data connection line.

7. The electronic device according to claim 1, wherein the pixel comprises a micro-LED.

8. The electronic device according to claim 1, further comprising a scan line and a data line, disposed in the display area.

9. The electronic device according to claim 1, wherein the display panel further comprises another data connection line, another pixel, and another scan connection line, the another data connection line is electrically connected between the data signal driving circuit and the another pixel, and the another scan connection line is electrically connected between the scan signal driving circuit and the another pixel, and wherein the data connection line and the scan connection line are included by a first angle, and the another data connection line and the another scan connection line are included by a second angle different from the first angle.
